

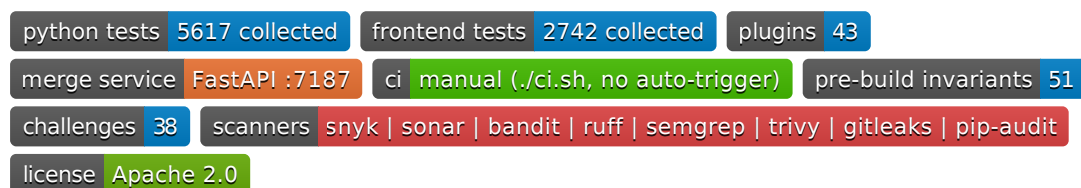
README



Боба / Boba

Multi-tracker meta-search for qBittorrent — self-hosted, containerised, private-tracker-aware.

[Quick start](#) · [Tokens & keys](#) · [Documentation](#) · [Architecture](#) · [Testing](#) · [Contributing](#)



Features

- **Merge Search Service** — FastAPI service (:7187) that fans out across 40+ trackers, deduplicates results, streams via SSE.
- **Real-time results** — `result_found` events arrive as each tracker completes, no blocking.
- **Private-tracker bridge** — Authenticated downloads via `webui-bridge.py` for RuTracker, Kinozal, NNM-Club, IPTorrents.
- **Freeleech-only IPTorrents** — Automation never costs ratio (see [constitution VIII](#)).

- **Opt-in quality stack** — SonarQube + Snyk + Semgrep + Trivy + Gitleaks + bandit + pip-audit behind docker-compose.quality.yml.
 - **Opt-in observability** — Prometheus + Grafana dashboards behind the same profile system.
 - **Hook system** — Register scripts via POST /api/v1/hooks for search / download events.
 - **Angular 21 SPA dashboard** — dark-themed, signals-based, per-tracker status chips with CAPTCHA re-login, virtual-scroll-ready sort.
 - **PWA-ready** — favicon + launcher icons + web manifest ship from frontend/public/.
 - **Container-first** — rootless Podman or Docker, single ./start.sh boot.
 - **Resource-pressure early-warning** — [challenges/scripts/resource_pressure_signature_challenge.sh](#) checks 5 independently-falsifiable host-pressure signatures (runaway RSS, thread-count vs ulimit -u, container EAGAIN cascades, swap pressure, cgroup memory pressure) so a forced-logout precursor is caught *before* user@1000.service gets SIGKILLED, not after. Wired into scripts/pre_build_verification.sh invariant 25 and a standing boba-resource-pressure-check.timer systemd-user unit. See [incident writeup](#).
-

Quick start

```
git clone https://github.com/milos85vasic/qBitTorrent.git
cd qBitTorrent
```

Optional: configure tracker credentials + tokens

```
cp .env.example .env
```

```
$EDITOR .env # see "Tokens & API keys" below
```

```
./setup.sh # one-time
```

```
./start.sh # Angular build + containers + plugin sync
```

Then visit:

URL	Purpose	Auth
http://localhost:7187/	Merge-search dashboard (SPA)	—
http://localhost:7187/docs	FastAPI Swagger UI	—
http://localhost:7187/openapi.json	OpenAPI schema	—
http://localhost:7186/	qBittorrent WebUI proxy	admin / admin

URL	Purpose	Auth
http://localhost:7185/	qBittorrent internal (container)	—
http://localhost:7188/	WebUI bridge (private-tracker downloads)	—

Tokens & API keys

⚠ **Configure any credentials you need in `.env` before running `./start.sh`.** The single source of truth is [docs/TOKENS_AND_KEYS.md](#) — which variable is mandatory, which is optional, and **where to register** for each one.

Tracker credentials — env vars AND cookies files (auto-loaded)

Boba supports **two mechanisms** for tracker auth, and both are respected together:

1. **Env vars in `.env`** — `<TRACKER>_USERNAME` / `<TRACKER>_PASSWORD` / `<TRACKER>_COOKIES`.
2. **Cookies files in `~/Downloads/`** — canonical naming `cookies_<tracker>.txt` (lowercase), Netscape TSV format exported from a logged-in browser session.

The loader `scripts/load-tracker-cookies.sh` auto-invokes before every `boba-svc up`, `boba-svc restart`, `install.sh` Stage 6, and `start.sh` boot — so refreshing a session is a one-step *browser re-export* → *restart*. Source directory defaults to `$HOME/Downloads`, overridable via `TRACKER_COOKIE_DIR` or `--dir <path>`.

See the **[Tracker Credentials Manual](#)** for the per-tracker matrix, browser-export step-by-step, precedence rules, and the security posture (§11.4.10.A leak audit, cookie-value-never-logged). Common questions: [FAQ](#).

Downloads and directories the system creates are owned by the account that started it — no `chown` step after every download. See the **[File Ownership Guide](#)** for why that requires `PUID=0` under rootless Podman, how to verify it on your own machine, and how to repair content created before the fix.

At a glance

Category	Mandatory	Optional	Documentation
qBittorrent WebUI	—	WEBUI_USERNAME, WEBUI_PASSWORD	§1
Private trackers	per-tracker	—	§2
Public tracker uplift	—	JACKETT_API_KEY, ...	§3
Metadata enrichment	—	TMDB_API_KEY, TVDB_API_KEY, ...	§4
Security scanning	—	SNYK_TOKEN, SONAR_TOKEN, ...	§5
Observability	—	GRAFANA_USER, GRAFANA_PASSWORD	§6
Orchestrator tuning	—	ALLOWED_ORIGINS, MAX_CONCURRENT_TRACKERS, ...	§7

Where to register (fast links)

Provider	Signup	Notes
RuTracker	https://rutracker.org/forum/register.php	Username + password
Kinozal	https://kinozal.tv/signup.php	May require invite
NNM-Club	https://nnmclub.to/forum/ucp.php?mode=register	Cookie-based auth
IPTorrents	https://iptorrents.com/	Invite-only; freeleech enforced
Jackett	https://github.com/Jackett/Jackett	Self-hosted indexer
TMDb	https://www.themoviedb.org/signup	v3 auth API key
TVDb	https://thetvdb.com/api-information	Subscription API
MusicBrainz	https://musicbrainz.org/doc/MusicBrainz_API	Free — needs UA only
Snyk	https://app.snyk.io/	Account → Auth Token
SonarCloud		

Provider	Signup	Notes
	https://sonarcloud.io/account/security/	Or run the compose SonarQube
Gitleaks	https://gitleaks.io/	Optional commercial license

Documentation

Everything the platform offers, indexed:

Tracked-Items + Status Documents

The mandated entry point (§11.4.57 / §11.4.212) for every tracker and status document in this repository — Last modified and Revision are sourced verbatim from each document's own revision header (§11.4.44); a — means the document currently carries no revision header or has no HTML/PDF export yet (never fabricated). Regenerated by `scripts/testing/update_readme_doc_links.sh` — never hand-typed (§11.4.57).

Document	Last modified	Revision
Issues	2026-08-21T19:41:44Z	40
Issues_Summary	—	—
Fixed	2026-08-21T19:41:44Z	25
Fixed_Summary	—	—
CONTINUATION	2026-08-20T12:17:48Z	27
PORTING-FROM-LAVA	2026-07-01T16:18:43Z	1
REMAINING_WORK_PLAN	2026-08-07T19:10:54Z	2
COMPLETION_STATUS	2026-06-06T00:00:00Z	1
RELEASE_READINESS_20260616	2026-06-16T10:30:00Z	1
QA_DISCOVERY_LEDGER	2026-08-21T15:05:00Z	16
GOVERNANCE_AUDIT_2026-08-07	2026-08-07T19:10:54Z	1
GOVERNANCE_AUDIT_2026-08-08_ROUND2	2026-08-09T12:54:30Z	6
browser_extension/Status	2026-08-20T12:27:23Z	16
browser_extension/Status_Summary	2026-08-20T12:27:23Z	4
browser_extension/RELEASE_READINESS	2026-06-13T13:10:00Z	4
codegraph/Status	2026-08-18T13:33:41Z	2
codegraph/Status_Summary	2026-08-20T12:27:23Z	1
features/Status	2026-08-21T00:00:00Z	10

Document	Last modified	Revision
features/Status_Summary	2026-08-20T12:27:23Z	8

Getting started & operation

- [docs/README.md](#) — secondary index of every file directly under docs/
- [docs/USER_MANUAL.md](#) — end-user walkthrough
- [docs/TOKENS_AND_KEYS.md](#) — **credentials / tokens / env vars (mandatory vs optional + registration links)**
- [docs/DESIGN_SYSTEM.md](#) — runtime-switchable palette catalogue (8 palettes, dark + light, Darcula default)
- [docs/PLUGINS.md](#) — the 43 plugin engines
- [docs/PLUGIN_TROUBLESHOOTING.md](#) — what to check when a plugin breaks
- [docs/MAGNET_LINKS.md](#) — magnet link implementation guide
- [docs/DOWNLOAD_FIX.md](#) — RuTracker plugin download fix
- [docs/RELEASE_TORRENT_UPLOAD_FIX.md](#) — release report: torrent file upload fix
- [docs/JACKETT_INTEGRATION.md](#) — Jackett integration
- [docs/DEAD_TRACKERS_EXPLAINED.md](#) — dead public tracker status
- [docs/TRACKER_AUTH_TESTING.md](#) — tracker credentials live testing
- [docs/MERGE_SEARCH_DIAGNOSTICS.md](#) — merge-search diagnostics + rebuild/restart contract

Architecture & subsystems

- [docs/architecture/](#) — 5 Mermaid diagrams (topology, search lifecycle, plugin execution, bridge, shutdown)
- [docs/DATA_MODEL.md](#) — Pydantic schemas & lifecycle (no relational DB)
- [docs/CONCURRENCY.md](#) — asyncio semaphore + TTL caches + retry/backoff
- [docs/OBSERVABILITY.md](#) — Prometheus + Grafana + OpenTelemetry
- [docs/PERFORMANCE.md](#) — benchmark + load + stress test layout
- [docs/api/openapi.json](#) — frozen OpenAPI spec (diffed in CI)
- [docs/API_Summary.md](#) — API domain summary
- [docs/Architecture_Summary.md](#) — architecture domain summary
- [docs/CODEGRAPH.md](#) — CodeGraph code-intelligence integration (also indexed above in Tracked-Items via codegraph/Status)
- [docs/CodeGraph_Summary.md](#) — CodeGraph domain summary
- [docs/MULTITRACK.md](#) — multi-track development registration (Track 11 rationale)
- [docs/CROSS_APP_THEME_PLAN.md](#) — cross-app theme plan

Quality & security

- [docs/SECURITY.md](#) — threat model + credential storage

- [docs/SCANNING.md](#) — Snyk + Sonar + Semgrep + Trivy + Gitleaks + bandit + pip-audit
- [docs/QUALITY_STACK.md](#) — the opt-in docker-compose.quality.yml stack
- [docs/AGENT_GUARDRAILS.md](#) — §11.4.109 anti-forgetting enforcement (subagent constitutional preamble + orchestrator pre-action checklist)
- [docs/CONSTITUTION_ADDENDUM_QUALITY.md](#) — constitution addendum: quality stack (Principle I exception)
- [docs/PLUGIN_AUDIT.md](#) — plugin audit matrix
- [docs/TEST_RESULTS.md](#) — snapshot of last full test run
- [docs/TEST_SUITE_GUIDE.md](#) — test suite guide
- [docs/testing/](#) — DDoS resilience + test-type matrix notes

Incidents & QA evidence

Per §11.4.238, automated HelixQA coverage is the discovery layer — manual/operator/agent-found defects are confirmation-only, and every out-of-band finding gets a coverage-escape-audit entry here.

- [docs/QA_DISCOVERY_LEDGER.md](#) — the discovery-channel ledger + coverage-escape schema (also indexed above in Tracked-Items)
- [docs/incidents/](#) — host-safety + CONST-033 forensic investigations, including the latest [2026-08-18 2nd perceived forced-logout](#) triage (root cause: partial, §11.4.6 UNCONFIRMED: boundary honestly stated; preventive gate: the resource-pressure signature challenge above)
- [docs/qa/](#) — per-item captured machine evidence (§11.4.5 / §11.4.69 / §11.4.83), one directory per workable item / task run — e.g. [docs/qa/BOB-116/](#)
- [docs/history/BOB-079-attributed-auto-commit-history.md](#) — retroactive, evidenced attribution for the 14 unattributed Auto-commit/sync: commits in v1.0.0-rc..HEAD (BOB-079/RD2-12). Attribution is read from each commit's DIFF, never its subject line; 3 of 14 are evidence-backed and 11 are recorded literally as UNKNOWN (§11.4.6 — a fabricated attribution is worse than an absent one). A forward-only record, never a history rewrite (§11.4.113)
- [docs/HOST_POWER_MANAGEMENT.md](#) — CONST-033 host power management hard ban
- [docs/guides/forced-logout-flight-recorder.md](#) — the flight-recorder guide for perceived-forced-logout triage

Research, planning, migration & session archives

Working documents, dated research notes, superseded-implementation plans, and dated session reports — browsable indexes rather than individually curated, since new dated entries land continuously and any per-file categorization here would drift the moment a new one is added.

- [docs/research/](#) — dated investigation write-ups (tracker discovery, plugin audits, browser-extension corpus)
- [docs/superpowers/](#) — long-horizon implementation plans + design specs
- [docs/browser_extension/](#) — architecture, changelog, install, store listing, user guide, coverage ledger, and internal [_analysis/_plan](#) working notes (Status docs indexed above in Tracked-Items)
- [docs/migration/](#) — Python-to-Go migration spec + parity gaps
- [docs/plans/](#) — dated implementation plans
- [docs/proposals/](#) — architectural proposals under consideration
- [docs/demos/](#) — dated demo write-ups
- [docs/issues/](#) — frozen post-mortems for specific incidents (distinct from the live docs/Issues.md tracker above)
- [docs/systemd/UNITS.md](#) — systemd unit reference
- [docs/SESSION_REPORT_2026-04-19.md](#) — rolling session report
- [docs/SYSTEMATIC_DEBUGGING_2026-04-19.md](#) — systematic debugging session log
- [docs/Demos_Summary.md](#) — demos domain summary
- [docs/Migration_Summary.md](#) — migration domain summary
- [docs/Scripts_Summary.md](#) — scripts domain summary
- [docs/Superpowers_Summary.md](#) — superpowers domain summary

Manual QA

Per §11.4.185, automated gates are necessary but not sufficient — every release deliverable requires a final human confirmation pass. Per-session manual-QA checklists live under [docs/manual-qa/](#), one file per session, in the §11.4.153 per-feature Status/video-confirmation format.

- [docs/manual-qa/2026-08-18-session-manual-qa-checklist.md](#) — 12-feature checklist covering the boba-stack reachability, cross-tracker search/dedup, download flow, BOB-112 /healthz DDoS mitigation, constitution v68 landing, resource-pressure timer, CONST-033 challenge, and pre-build gate for the 457cca4→63f3f88 session range

Testing

- [docs/TESTING.md](#) — catalogue of every test type (30 rows)
- [docs/COVERAGE_BASELINE.md](#) — per-module coverage gates

- [tests/README.md](#) — layout of the tests/ tree

Development

- [CONTRIBUTING.md](#) — contribution workflow
- [CLAUDE.md](#) — Claude Code agent protocol (TDD + rebuild-reboot)
- [AGENTS.md](#) — runtime development guidance
- [.specify/memory/constitution.md](#) — **binding architectural contract (v1.1.0)**
- [docs/scripts/](#) — per-script user guides (§11.4.18), one <name>.md per scripts/*.sh / challenges/scripts/*.sh — e.g. [commit-push-all.md](#), [capture-workable-items-db-delta.md](#), [regenerate-continuation-exports.md](#)

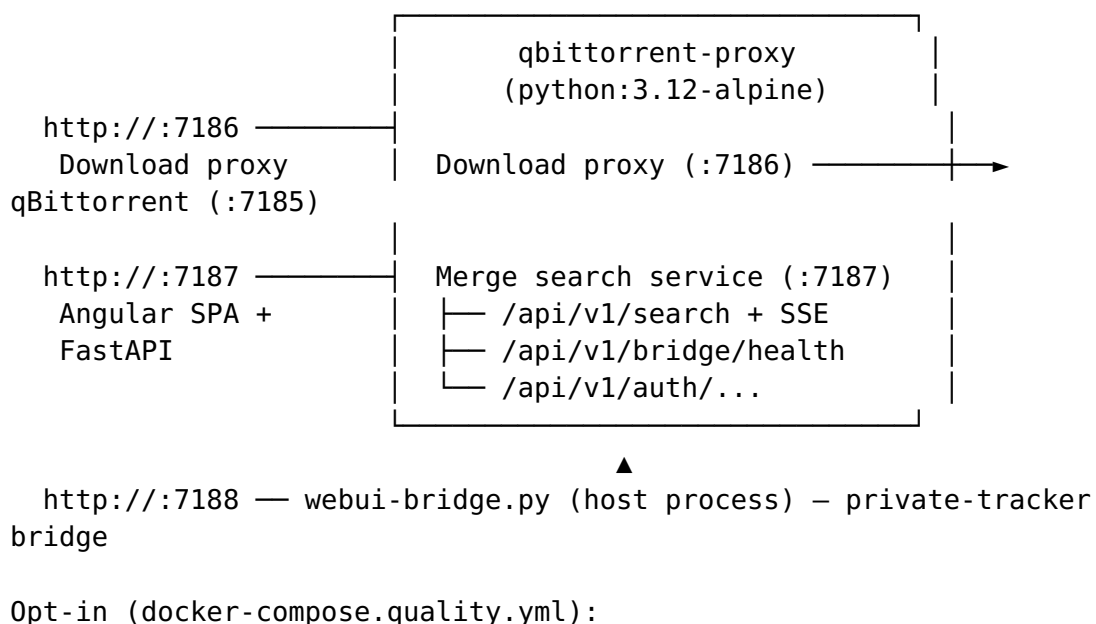
Courses (self-paced, Asciiinema)

- [courses/01-operator/](#) — Your first search
- [courses/02-plugin-author/](#) — Authoring a nova3 plugin
- [courses/03-contributor/](#) — TDD + rebuild-reboot deep dive
- [courses/04-security-ops/](#) — Threat model + scanner bundle

Release / CI

- [CHANGELOG.md](#)
 - [releases/README.md](#) — release artefacts layout
 - [docs/OUT_OF_SANDBOX.md](#) — items requiring external credentials (HelixQA / OpenCode / submodule orgs)
-

Architecture



http://:9000	SonarQube	profile: quality
http://:9090	Prometheus	profile: observability
http://:3000	Grafana	profile: observability
(run-once)	Snyk / Semgrep / Trivy / Gitleaks	profile: run-once

Full Mermaid renders in [docs/architecture/](#).

Testing

Non-live-HTTP suites – run anywhere, ~10s

```
python3 -m pytest tests/unit/ tests/e2e/ tests/contract/ --no-cov -q
```

Benchmarks – dedupe perf + live-HTTP fan-out benchmarks, ~4min

```
python3 -m pytest tests/benchmark/ --no-cov -q
```

Security – hits the live merge service, needs stack up

```
python3 -m pytest tests/security/ --no-cov -q
```

Integration – same

```
python3 -m pytest tests/integration/ --no-cov -q -m "not requires_credentials"
```

Frontend

```
npm --prefix frontend ng test --watch=false
```

Full scanner sweep (non-interactive; skips scanners with missing tokens)

```
./scripts/scan.sh --all
```

Pre-build invariant sweep – 25 checks (workable-items integrity, docs-chain

sync, resource-pressure signature, mutation-residue, §11.4.238 discovery

ledger freshness, ...) run before every build

```
./scripts/pre_build_verification.sh
```

HelixQA Challenges – 31 scripts under challenges/scripts/, e.g.:

```
./challenges/scripts/resource_pressure_signature_challenge.sh
```

```
./challenges/scripts/ddos_resilience_challenge.sh --healthz
```

See [docs/TESTING.md](#) for the 30-row test-type catalogue.

Releases

Non-interactive builder at [scripts/build-releases.sh](#) produces artefacts under [releases/<version>/](#):

```
./scripts/build-releases.sh          # all targets, all  
                                     channels  
./scripts/build-releases.sh frontend # one target  
./scripts/build-releases.sh --channel release
```

Each artefact ships with SHA256SUMS + BUILD_INFO.json.

Contributing

See [CONTRIBUTING.md](#) and the TDD protocol in [CLAUDE.md](#). PRs must keep the following green:

- Python unit + e2e + contract (pytest — 4576 tests collected, see docs/TESTING.md)
 - Frontend Vitest (ng test — 2742 tests collected, see docs/TESTING.md)
 - Ruff + bandit + shellcheck (via scripts/scan.sh)
-

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